

Juan Antonio Monleón de la Lluvia

Nuclear Engineer · PhD in Nuclear Data & Neutronics

CONTACT

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EDUCATION

PhD in Nuclear Data & Neutronics
Université Paris-Saclay · ED
PHENIICS 2023-2026

**Double Master in Industrial Engineering
and Nuclear Science and Technology**
Polytechnic University of
Madrid 2021-2023

**Master in Management and Direction of
SMEs**
EFEM Business School 2024-2025

**Bachelor in Industrial Technology
Engineering**
Polytechnic University of
Madrid 2017-2021

TECHNICAL SKILLS

Nuclear Codes: MCNP, Serpent, OpenMC,
NJOY, ADVANTG
Programming: Python, Git, Claude Code
Documentation: LaTeX, Microsoft 365

ABOUT ME

Nuclear engineer working on nuclear data and on the sensitivity and uncertainty analysis of neutronic simulations, from the evaluation of covariances to their propagation in shielding and reactor-dosimetry calculations. Proactive and quick to adapt to new codes, tools and problem domains.

RESEARCH EXPERIENCE

PhD in Nuclear Data & Neutronics

ASNR, France · Université Paris-Saclay Nov 2023 - Oct 2026

Thesis: “Nuclear-Data Uncertainty Quantification for PWR Vessel Aging with a ⁵⁶Fe Elastic-Scattering Evaluation” (defended 1 October 2026)

- Sensitivity analysis and Monte Carlo sampling to propagate nuclear-data uncertainties; re-evaluated ⁵⁶Fe elastic angular distributions and covariances from EXFOR data
- Applied variance-reduction techniques (ADVANTG weight windows) to improve efficiency in deep-penetration shielding calculations
- Supervised a Master's student (Mar–Sep 2025) on nuclear-data uncertainty quantification in PWR source-term calculations

Master's Thesis Research (MII)

UPM, Spain Jun 2023 - Oct 2023

- Developed “Enhancing the EXFOR nuclear data library with Machine Learning Techniques”
- Implemented outlier detection methods for proton-induced reactions

Research Internship — Master's Thesis (MUCTN)

SCK CEN, Belgium Sep 2022 - Jun 2023

- Completed Master's thesis titled “Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques”
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

Master Internship

ENUSA Advanced Industries, Spain Oct 2021 - Apr 2022

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

PUBLICATIONS

- J. A. Monleon de la Lluvia, M. Brovchenko, D. Rochman, E. Dumonteil, “Towards Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations”, *Nuclear Science and Engineering*, 2025. DOI: 10.1080/00295639.2025.2510048 · pre-print 2025

LANGUAGES

Spanish Native language

English C2 Level

French C1 Level

AWARDS

Best Academic Report for Nuclear Master
Polytechnic University of Madrid (2022-2023)

Best Master Thesis Award
SCK CEN, Belgium (2023)

Best Poster Award
Journées des Thèses of IRSN (2024) - [View Poster](#)

COMMUNITY

JEFF Community
Active member · Joint Evaluated Fission and Fusion File (OECD/NEA)

INDEN Community
Member · International Nuclear Data Evaluation Network (IAEA)

SINBAD Community
Member · Shielding Integral Benchmark Archive & Database (OECD/NEA)

REFERENCES

Dr. Mariya Brovchenko
ASNR, France
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Dr. Dimitri Rochman
Nagra, Switzerland
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Prof. Óscar Cabellos
Universidad Politécnica de Madrid, Spain
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CONFERENCES & PRESENTATIONS

FIRST AUTHOR	
• “Evaluation of elastic angular distribution covariances for ⁵⁶Fe from EXFOR experimental data” — WONDER 2026, Aix-en-Provence	Jun 2026
• “Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to ⁵⁶Fe” — PHYSOR 2026	Apr 2026
• “Nuclear Data Uncertainty Quantification Using the VENUS-3 Benchmark Experiment” — ND 2025, Madrid	Jun 2025
• “Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing-like problems” — ANS Winter Conference RPSD, Orlando	Nov 2024
CO-AUTHORED	
• “Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6” — PHYSOR 2026	Apr 2026
• “Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations” — PHYSOR 2026	Apr 2026

SOFTWARE & TOOLS

KIKA — Desktop Workspace for Nuclear Data	
Author & lead developer	2025 - Present
• Opens ENDF, ACE, EXFOR and SDF files, drives a local NJOY installation, reads MCNP and Serpent runs, and carries the results into nuclear-data sampling, sensitivity and uncertainty analysis, all in a single desktop application	
• Also author of kika-nd , the open-source Python library (PyPI) I wrote to provide the computational core behind the application	

TRAINING

- OpenMC Course — NEA, Paris	Dec 2025
- NJOY Course — NEA, Paris	Dec 2024
- Sensitivity Analysis Summer School (SAMO) — UNIPR, Parma	Jun 2024
- Serpent Bootcamp — NEA, Paris	Nov 2023
- Course in "Nuclear Data for Energy and Non-Energy Applications" — GREAT PIONEER	Dec 2022